

Payment Service API Documentation

API Endpoints

The Payment Service provides the following RESTful API endpoints:

Customer Management

Endpoint	Method	Description
-----	-----	-----
`/payment/customer/:email`	GET	Get a customer by email
`/payment/customer`	POST	Create a new customer

Payment Method Management

Endpoint	Method	Description
-----	-----	-----
`/payment/payment-method`	POST	Add a new payment method to a customer
`/payment/payment-method-details`	POST	Add a payment method with detailed card information
`/payment/payment-method`	DELETE	Delete a payment method
`/payment/payment-methods/:customerId`	GET	List all payment methods for a customer

Payment Processing

Endpoint	Method	Description
----------	--------	-------------

-----	-----	-----
-------	-------	-------

<code>/payment/payment-intent`</code>	POST	Create a payment intent and process payment
---------------------------------------	------	---

<code>/payment/refund`</code>	POST	Process a refund for a payment
-------------------------------	------	--------------------------------

Request & Response Formats

All API endpoints accept and return JSON data. The standard response format includes:

```
```json
{
 "success": true/false,
 "data": { ... }, // For successful responses
 "message": "..." // For error responses
}
```
```

Authentication Methods

The Payment Service accepts authentication via Bearer token in the Authorization header:

```
```plaintext
Authorization: Bearer <token>
```
```

This token is passed to the Order Service when updating order status after successful payment.

API Examples

1. Create a Customer

****Request:****

``json

POST /payment/customer

Content-Type: application/json

```
{  
  "email": "customer@example.com",  
  "name": "John Doe"  
}  
``
```

****Response:****

``json

```
{  
  "success": true,  
  "data": {  
    "customerId": "cus_1234567890"  
  }  
}
```

```
}  
...
```

2. Get a Customer

****Request:****

```
``plaintext  
GET /payment/customer/customer@example.com  
...
```

****Response:****

```
``json  
{  
  "success": true,  
  "data": {  
    "customerId": "cus_1234567890"  
  }  
}  
...
```

3. Add a Payment Method

****Request:****

```
``json
```

POST /payment/payment-method

Content-Type: application/json

```
{
  "customerId": "cus_1234567890",
  "paymentMethodId": "pm_1234567890"
}
...
```

****Response:****

```
```json
{
 "success": true,
 "data": {
 "payment_method_id": "pm_1234567890",
 "card_last4": "4242",
 "card_brand": "visa"
 }
}
...
```

### 4. Add a Payment Method with Details

**\*\*Request:\*\***

```
```json
```

POST /payment/payment-method-details

Content-Type: application/json

```
{
  "customerId": "cus_1234567890",
  "paymentMethodId": "pm_1234567890",
  "cardholderName": "John Doe",
  "last4": "4242",
  "cardType": "visa",
  "expiryDate": "12/25",
  "isDefault": true
}
...
```

****Response:****

```
```json
{
 "success": true,
 "data": {
 "payment_method_id": "pm_1234567890",
 "card_last4": "4242",
 "card_brand": "visa",
 "cardholder_name": "John Doe",
 "expiry_date": "12/25",
 "is_default": true
 }
}
```

```
}
...
```

### ### 5. List Payment Methods

**\*\*Request:\*\***

```
``plaintext
GET /payment/payment-methods/cus_1234567890
...
```

**\*\*Response:\*\***

```
``json
{
 "success": true,
 "data": [
 {
 "payment_method_id": "pm_1234567890",
 "card_last4": "4242",
 "card_brand": "visa",
 "cardholder_name": "John Doe",
 "expiry_date": "12/25",
 "is_default": true
 }
]
}
```

```
...
```

### ### 6. Delete Payment Method

**\*\*Request:\*\***

```
``json
```

```
DELETE /payment/payment-method
```

```
Content-Type: application/json
```

```
{
 "customerId": "cus_1234567890",
 "paymentMethodId": "pm_1234567890"
}
```

```
...
```

**\*\*Response:\*\***

```
``json
```

```
{
 "success": true,
 "data": {
 "success": true,
 "message": "Payment method deleted successfully"
 }
}
```

```
...
```



### ### 7. Create Payment Intent

**\*\*Request:\*\***

``json

POST /payment/payment-intent

Content-Type: application/json

Authorization: Bearer <token>

```
{
 "amount": 2500,
 "customerId": "cus_1234567890",
 "paymentMethodId": "pm_1234567890",
 "orderId": "order_1234567890",
 "currency": "usd",
 "returnUrl": "https://example.com/checkout/complete"
}
``
```

**\*\*Response:\*\***

``json

```
{
 "success": true,
 "data": {
 "payment_intent_id": "pi_1234567890",

```

```
 "status": "succeeded",
 "amount": 2500,
 "currency": "usd"
 }
}
...
```

### ### 8. Process Refund

**\*\*Request:\*\***

``json

POST /payment/refund

Content-Type: application/json

```
{
 "paymentIntentId": "pi_1234567890",
 "amount": 1000,
 "reason": "customer_requested"
}
...
```

**\*\*Response:\*\***

``json

```
{
 "success": true,
```

```
"data": {
 "refund_id": "re_1234567890",
 "status": "succeeded",
 "amount": 1000,
 "currency": "usd",
 "payment_intent": "pi_1234567890"
}
}
...
```

## ## Error Responses

When an error occurs, the API returns an appropriate HTTP status code and a JSON response with error details:

```
``json
{
 "success": false,
 "message": "Error message describing what went wrong"
}
...
```

Common error status codes:

- `400` - Bad Request (missing or invalid parameters)
- `404` - Not Found (resource doesn't exist)
- `500` - Internal Server Error (server-side issue)